

On-Premises Active Directory to Microsoft Entra ID Synchronization using Microsoft Entra Connect

Objective

The objective of this project is to configure a hybrid identity environment by synchronizing on-premises Active Directory users and groups with Microsoft Entra ID using Microsoft Entra Connect. This enables centralized identity management, seamless authentication, and secure access to cloud services using on-premises credentials.

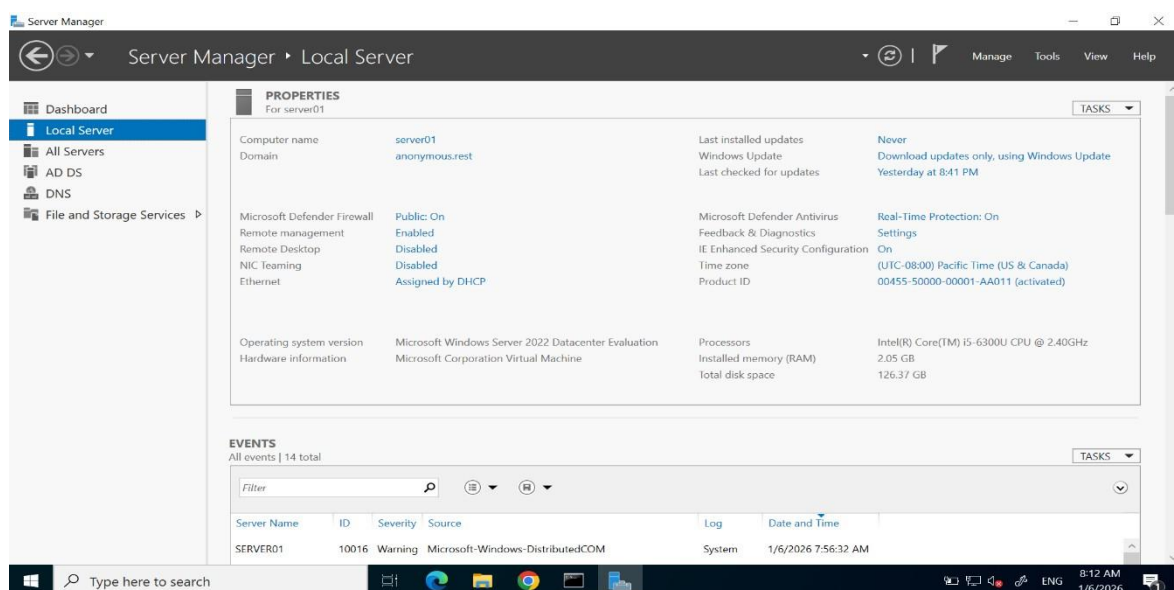
Tools & Technologies Used

- Windows Server 2022 (On-Premises Active Directory)
- Microsoft Entra ID (Azure AD)
- Microsoft Entra Connect Sync
- Azure Portal
- Active Directory Users & Computers (ADUC)

Deployment Steps

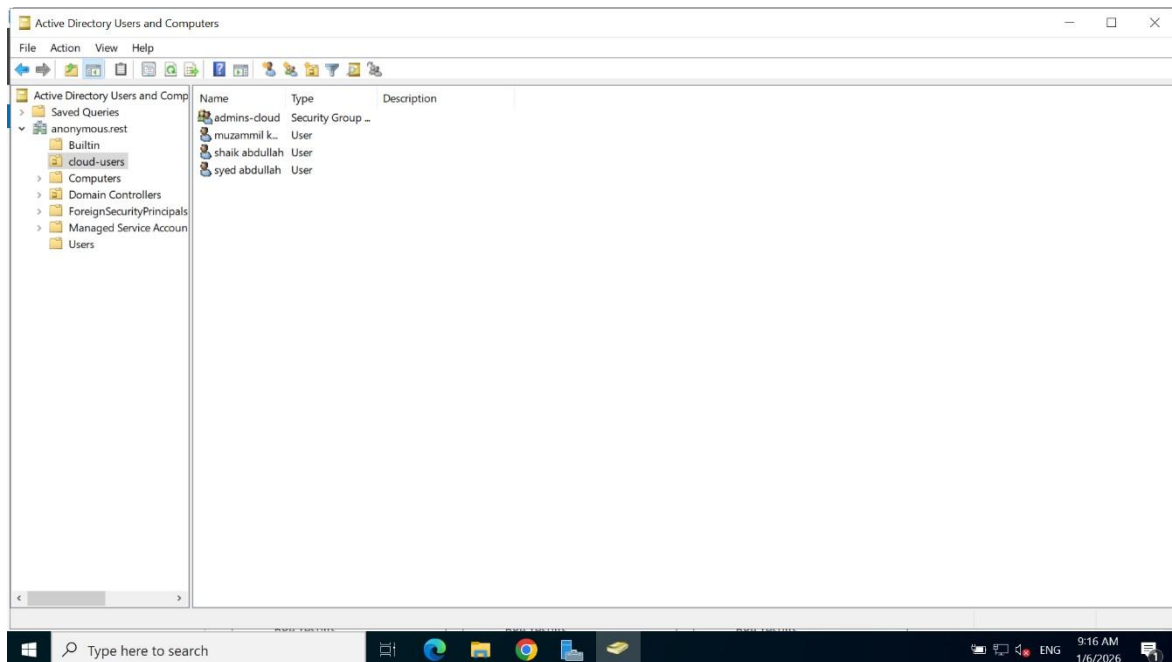
Step 1: Successfully Installed ADDS

I have successfully installed Active Directory and Domain Services on windows server named server01.



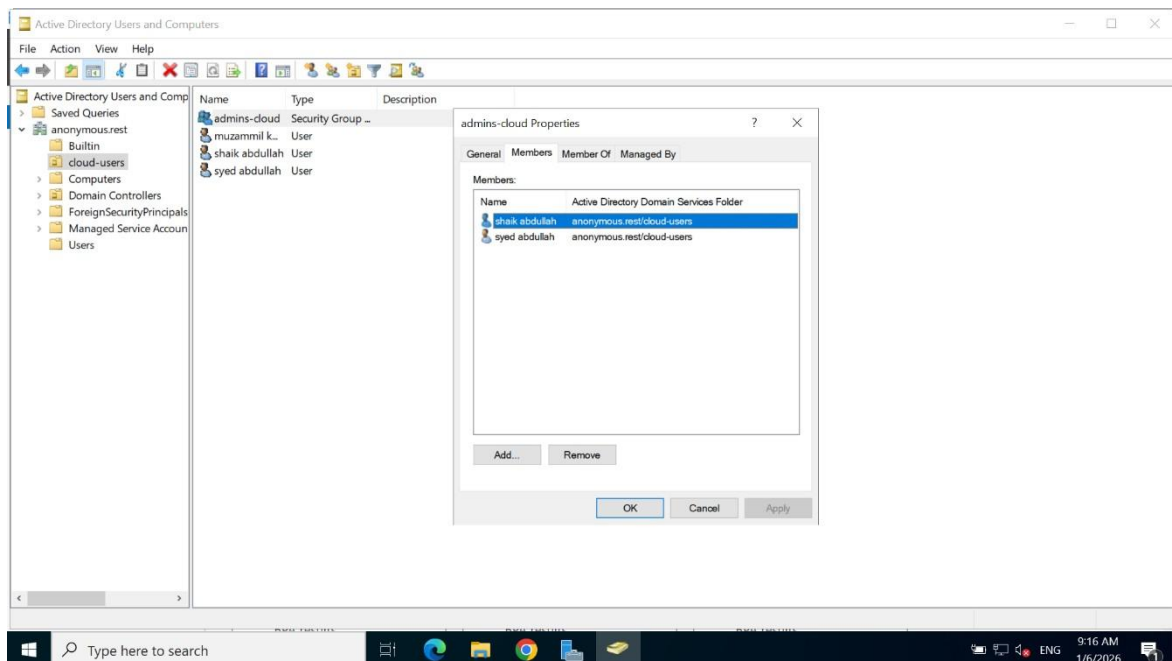
Step 2: Creation of OU , Users and Group

I have created an OU named Cloud-users in Active Directory Users and Computers and in that OU I have created a Security group and users.



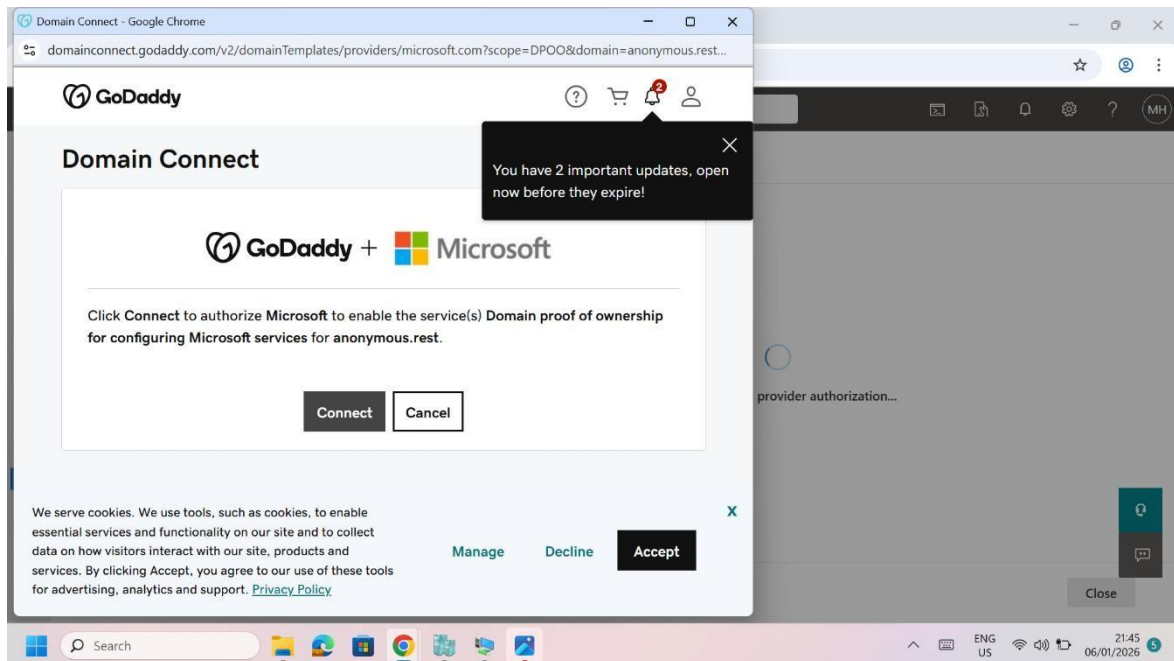
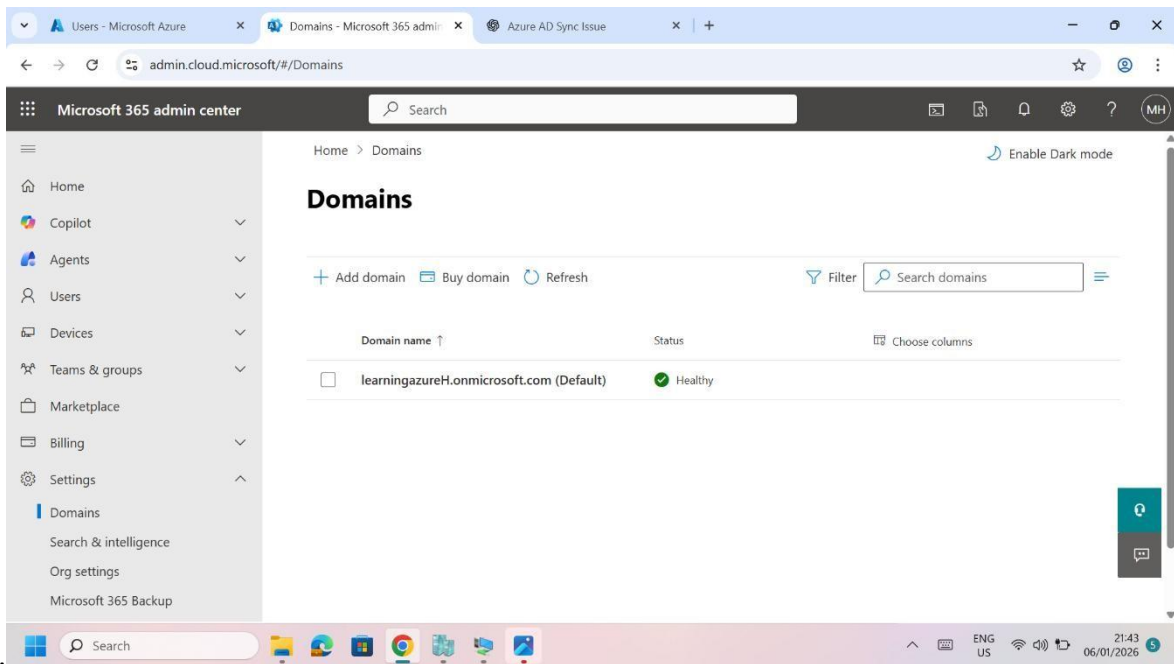
Step 3: Adding of members in group

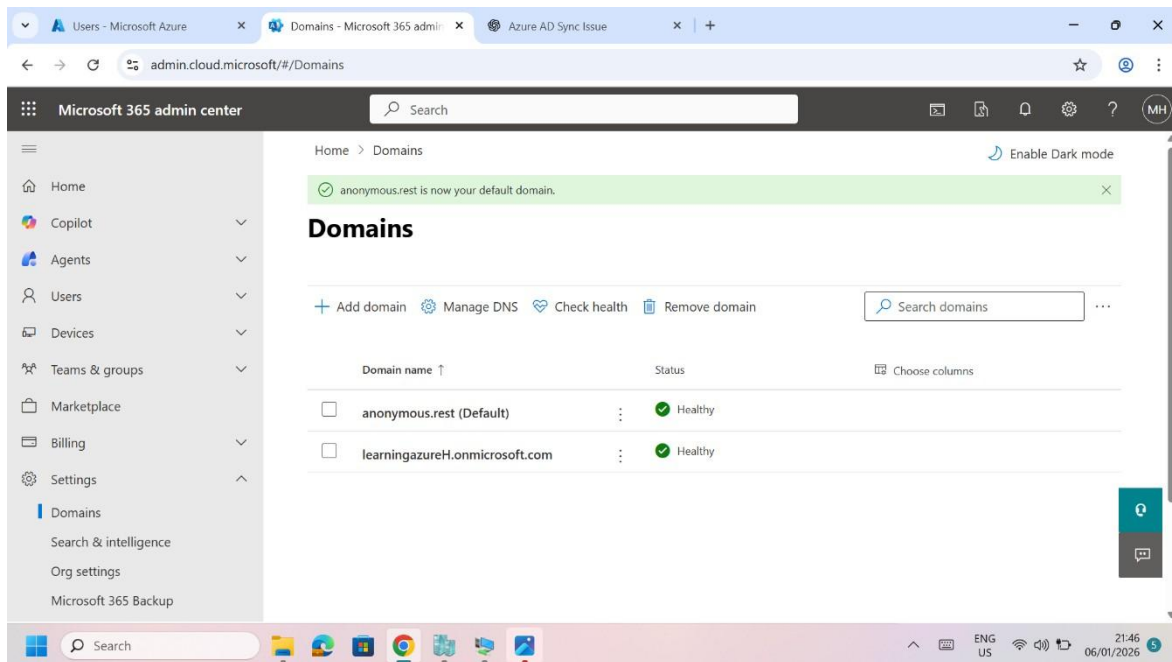
I have added two members into a group named admins-cloud.



Step 4: Adding and verifying domain

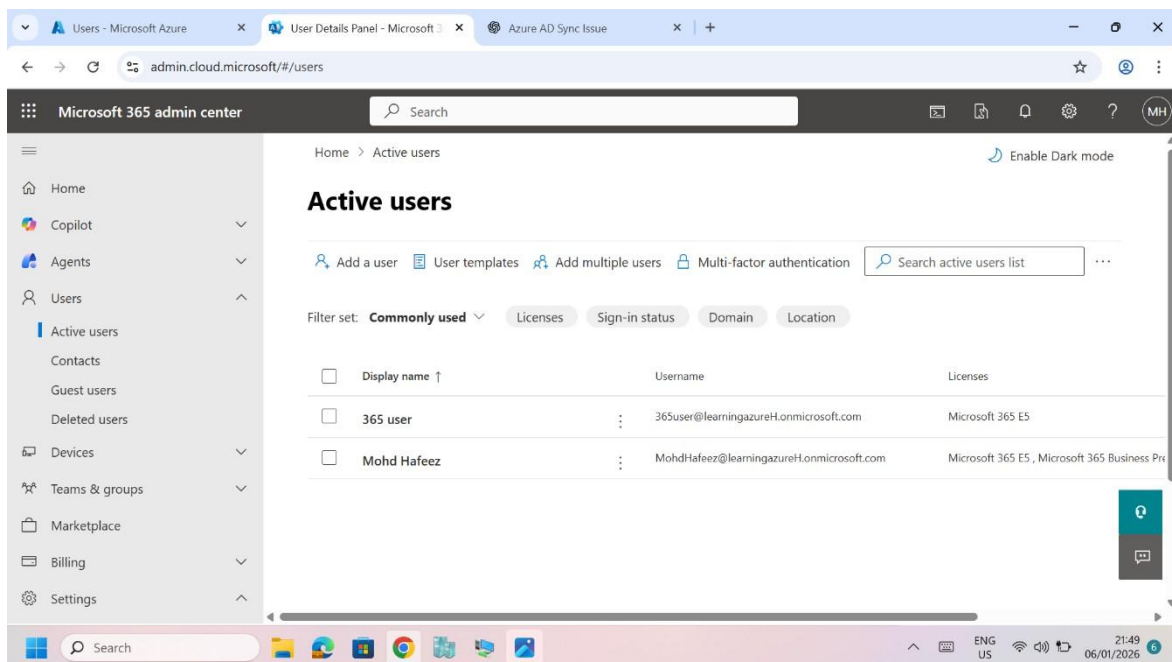
I have logged into admin.cloud.microsoft and add ,verify and configure my domain successfully and make it as a primary or default domain.





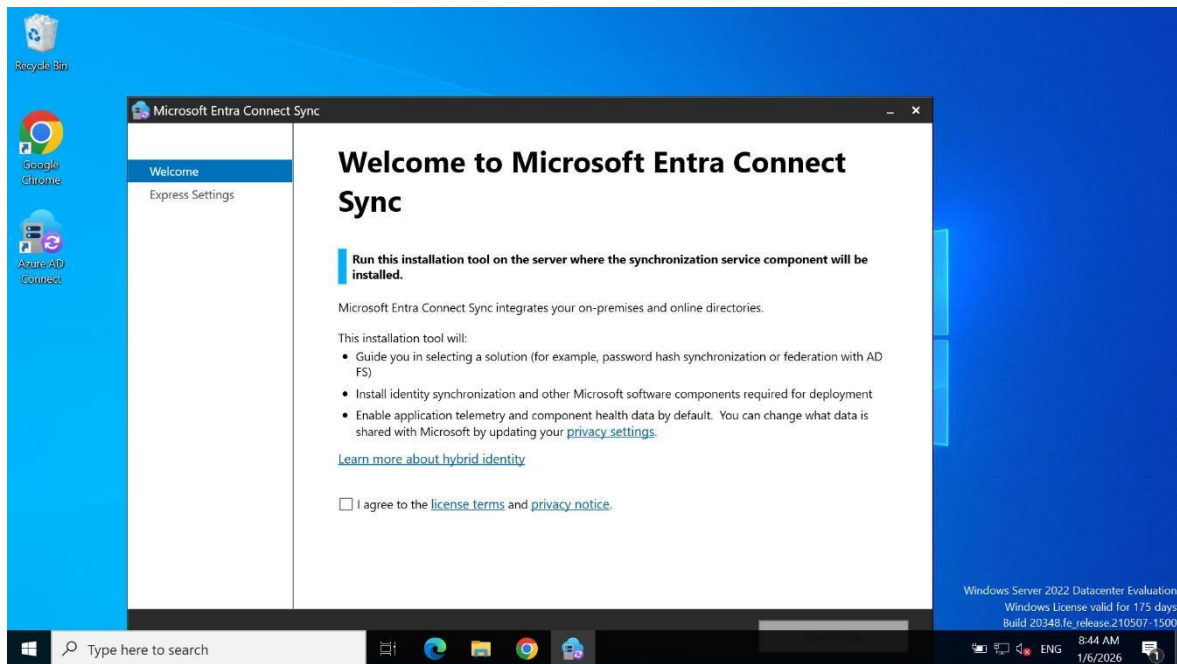
Step 5: Creation of a user and assigning a license

After successfully adding a domain I have created a user named 365user and assign is Microsoft 365 E5 license to that user.



Step 6: Launch Microsoft Entra Connect

Microsoft Entra Connect setup was launched on the on-premises Windows Server to start the hybrid identity configuration process.



Step 7: Select Custom Installation

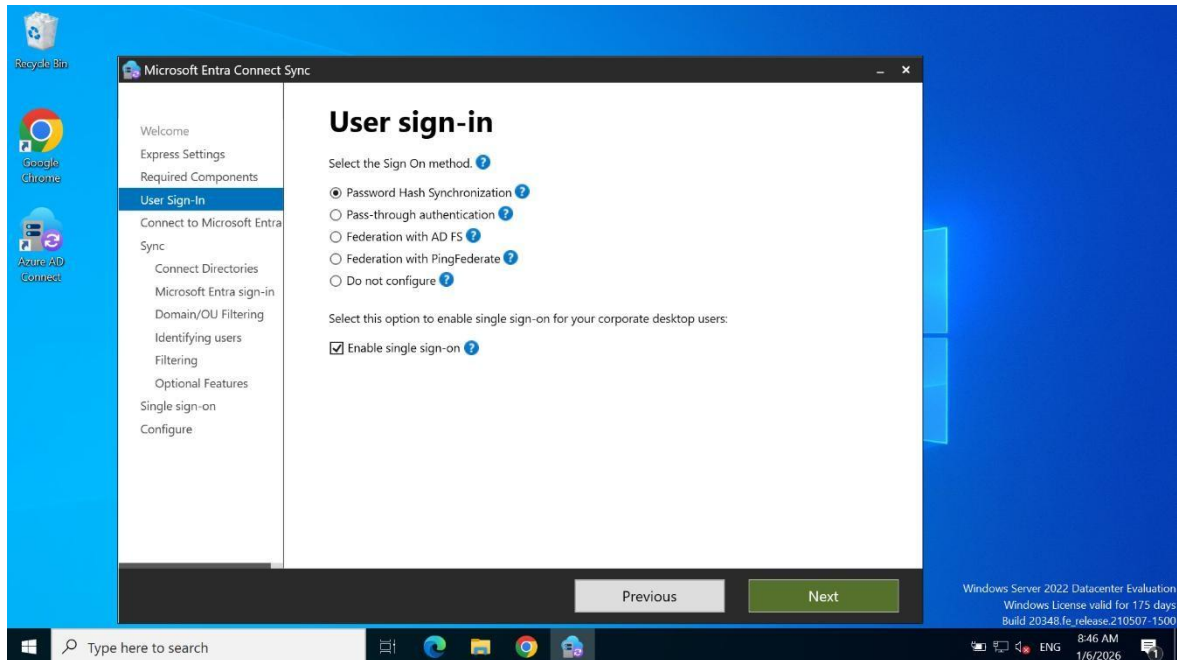
Custom installation was selected to manually control synchronization options, authentication method, filtering, and optional features.

Step 8: Review Required Components

The installer verified and prepared required components such as synchronization services and connectors needed for Entra Connect.

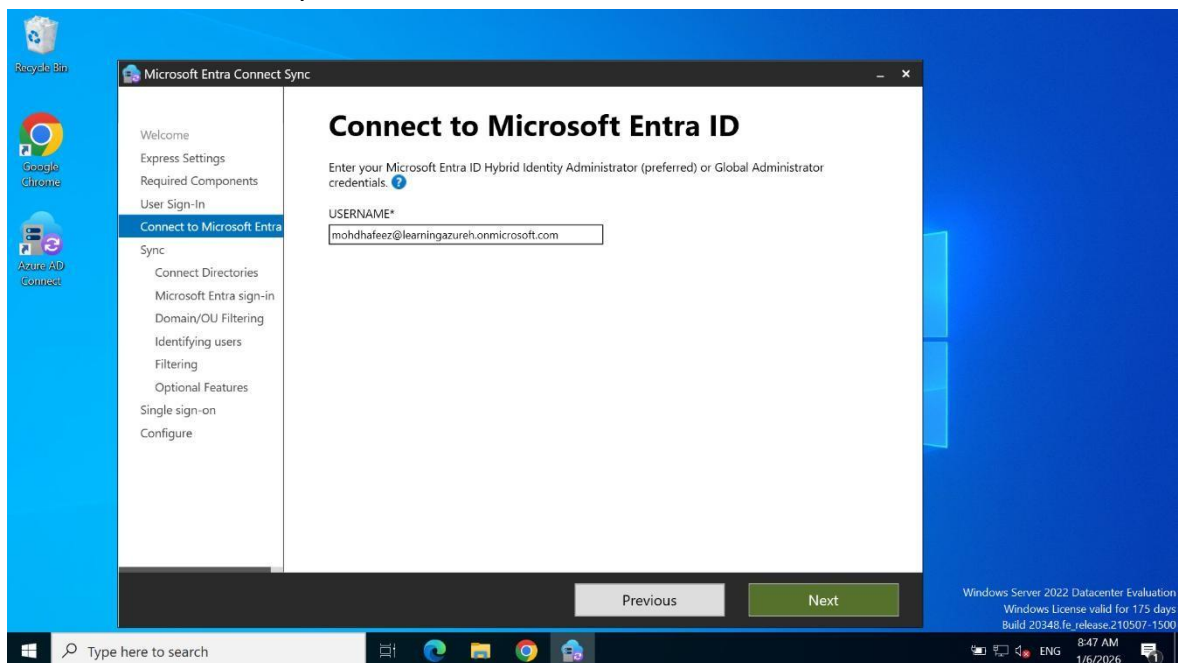
Step 9: Choose User Sign-In Method

Password Hash Synchronization (PHS) was selected as the sign-in method to allow users to authenticate to cloud services using synced password hashes. Single Sign-On was enabled to allow seamless authentication for domain-joined devices.



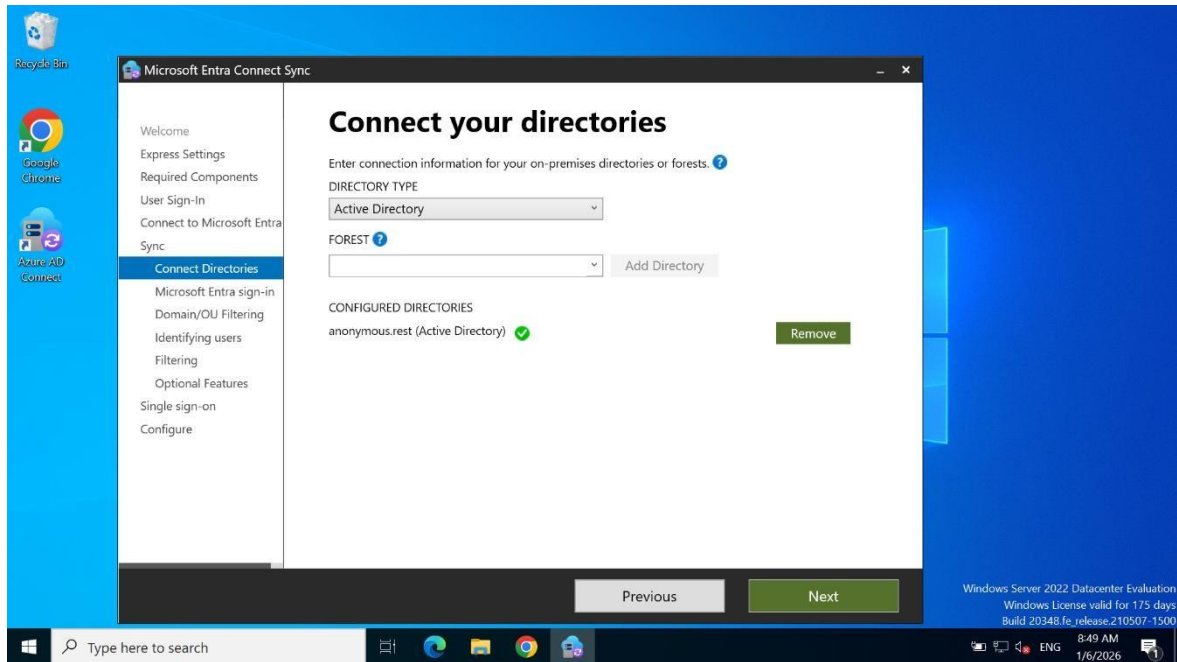
Step 10: Connect to Microsoft Entra ID

Microsoft Entra ID credentials (Hybrid Identity Administrator / Global Administrator) were used to establish a secure connection with the tenant and verified entra ID tenant connection successfully.



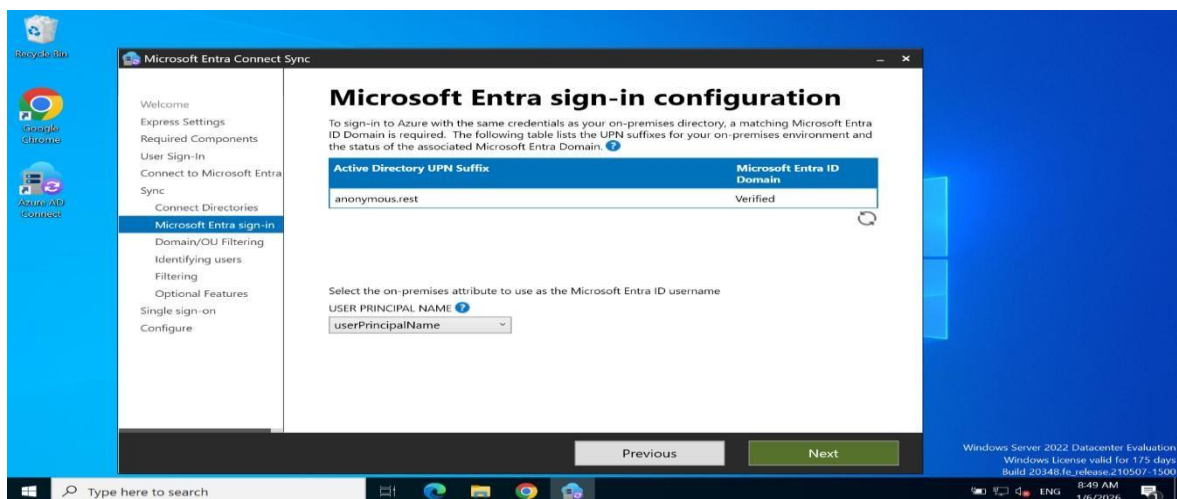
Step 11: Select Directory Type

Active Directory was selected as the directory type to synchronize identities from the on-premises environment and on-premises Active Directory forest (anonymous.rest) was added and authenticated successfully. The Active Directory forest was verified and showed a successful connection status.



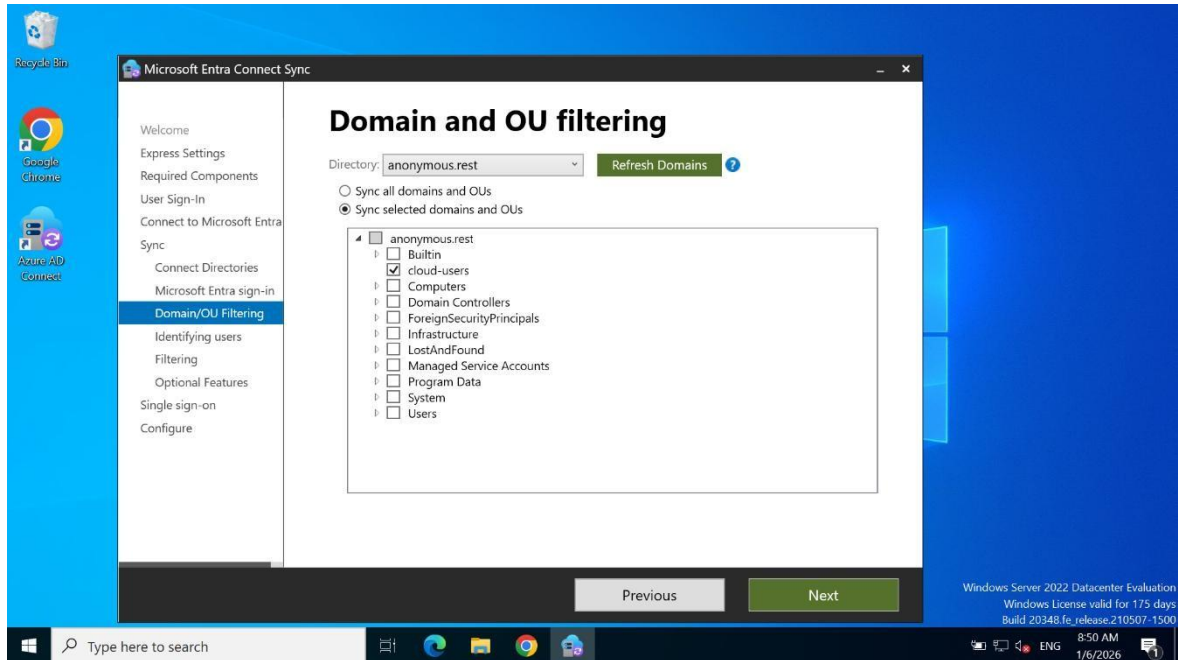
Step 12: Microsoft Entra Sign-In Configuration

UPN suffix mapping was validated to ensure on-premises user logins match Microsoft Entra ID domain names and UserPrincipalName (UPN) was selected as the attribute used for Microsoft Entra ID sign-in.



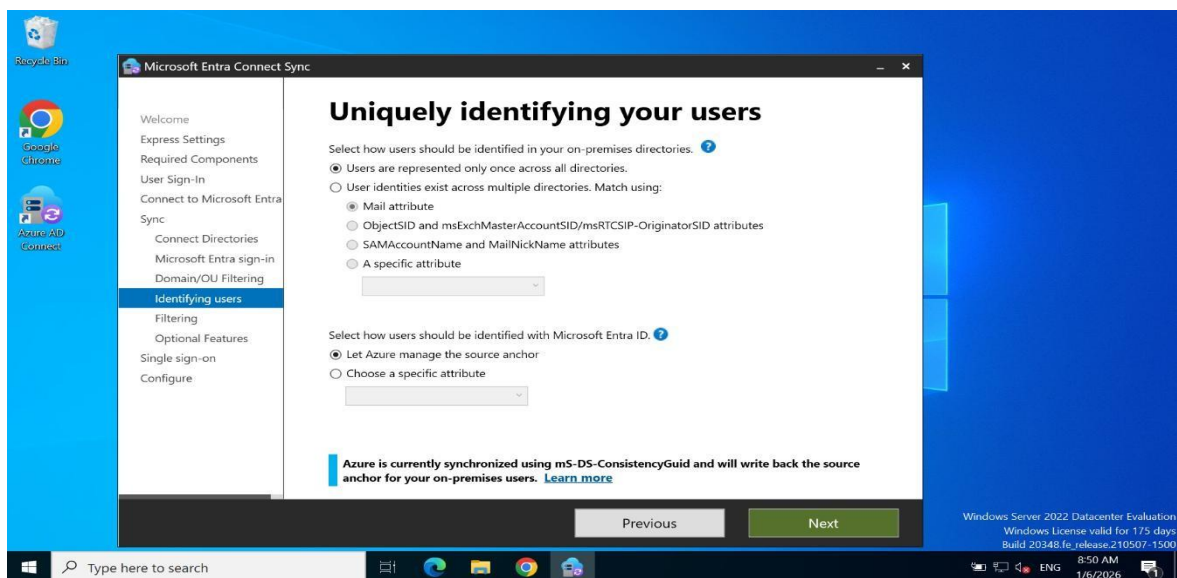
Step 13: Domain and OU Filtering Selection

The option to sync selected domains and OUs was chosen instead of syncing the entire directory. Only the required OU (cloud-users) was selected to control which users are synchronized to the cloud. Default OUs such as Built-in, Computers, and System were excluded to avoid unnecessary synchronization.



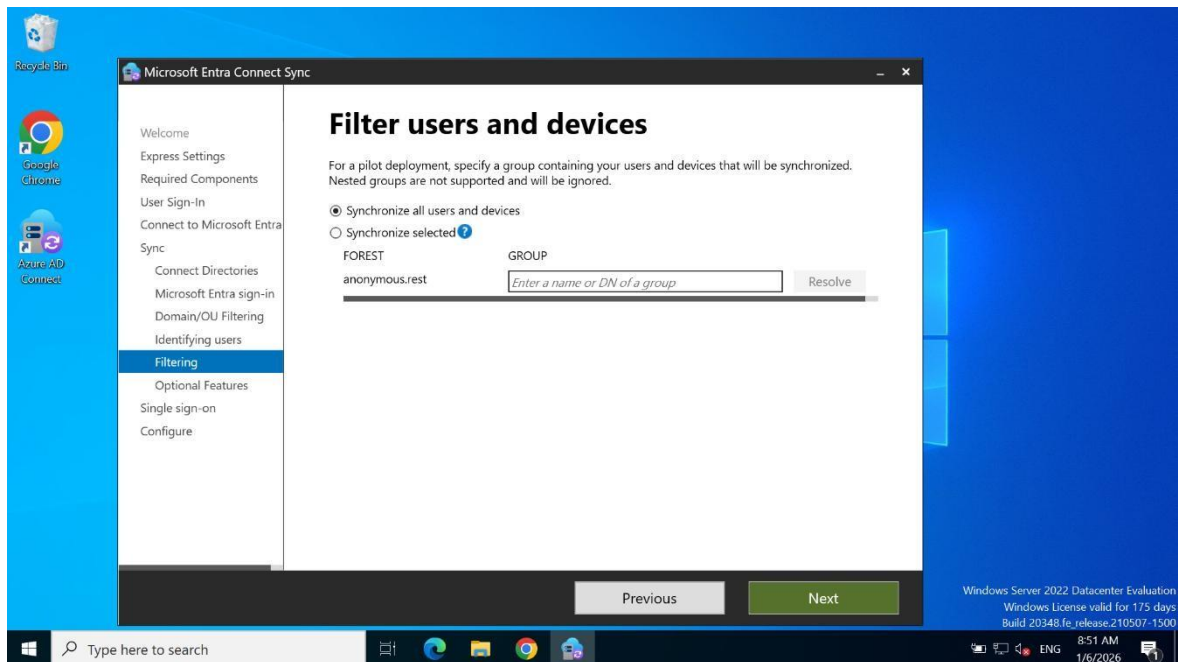
Step 14: Uniquely Identifying Users

The option “Users are represented only once across all directories” was selected to prevent duplicate identities and Azure was allowed to manage the source anchor automatically using the ms-DS-ConsistencyGuid attribute.



Step 15: User and Device Filtering

The option to synchronize all users and devices was selected for this deployment.

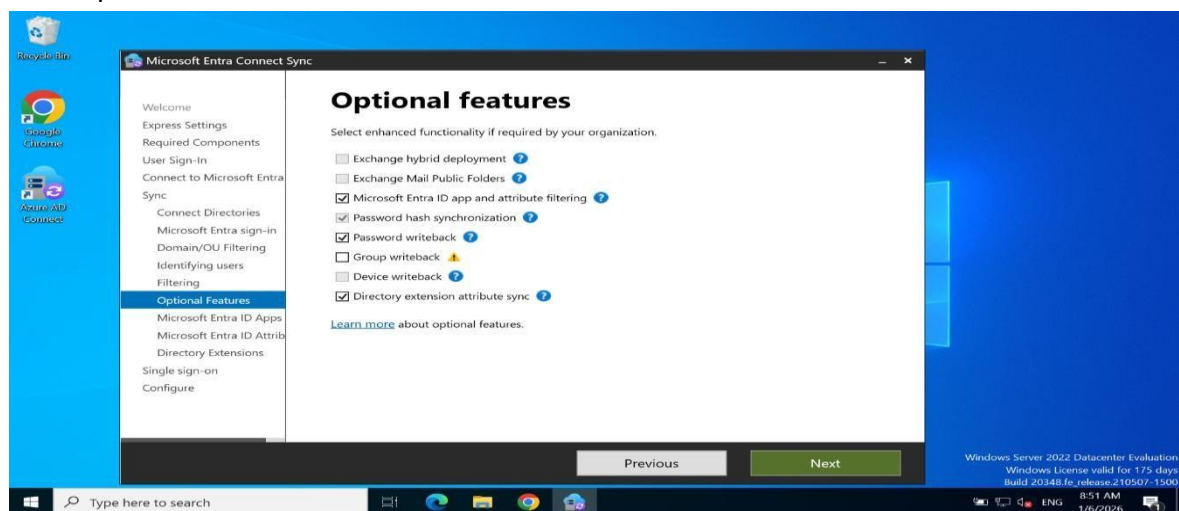


Step 16: Enable Password Hash Synchronization

Password Hash Synchronization was enabled to maintain password consistency between on-premises AD and Entra ID.

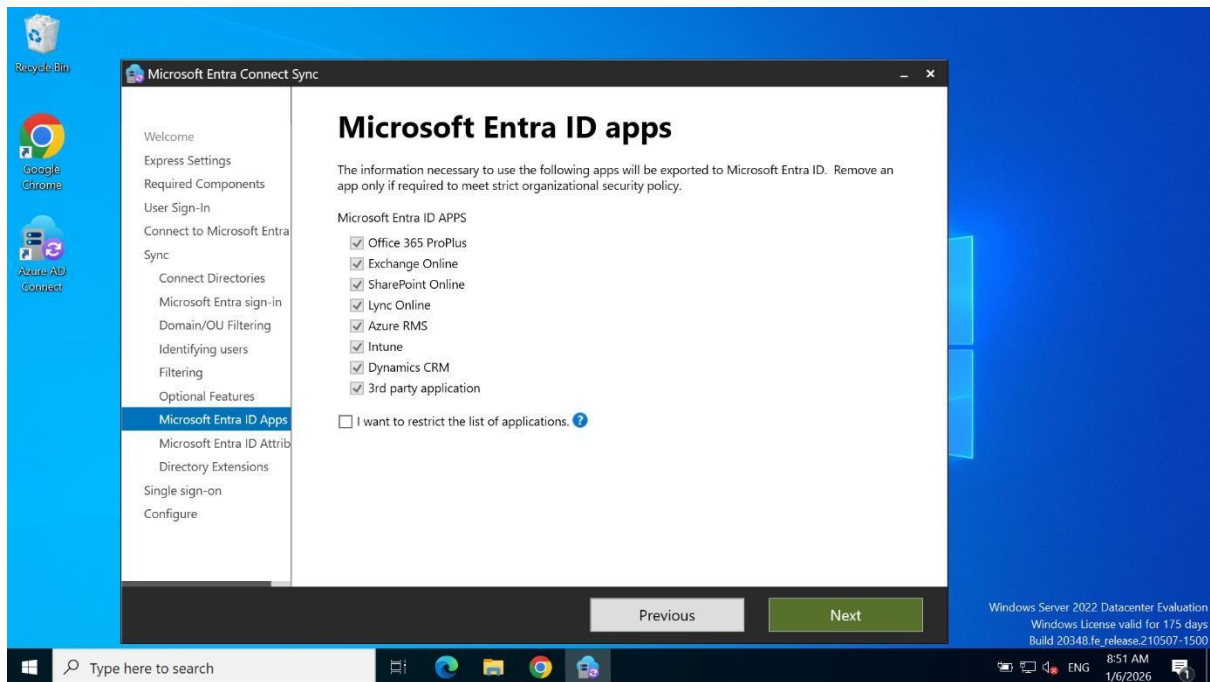
Password writeback was enabled to allow password changes in Entra ID to be written back to on-premises AD.

Directory extension attributes were enabled to synchronize additional AD attributes to Entra ID if required.



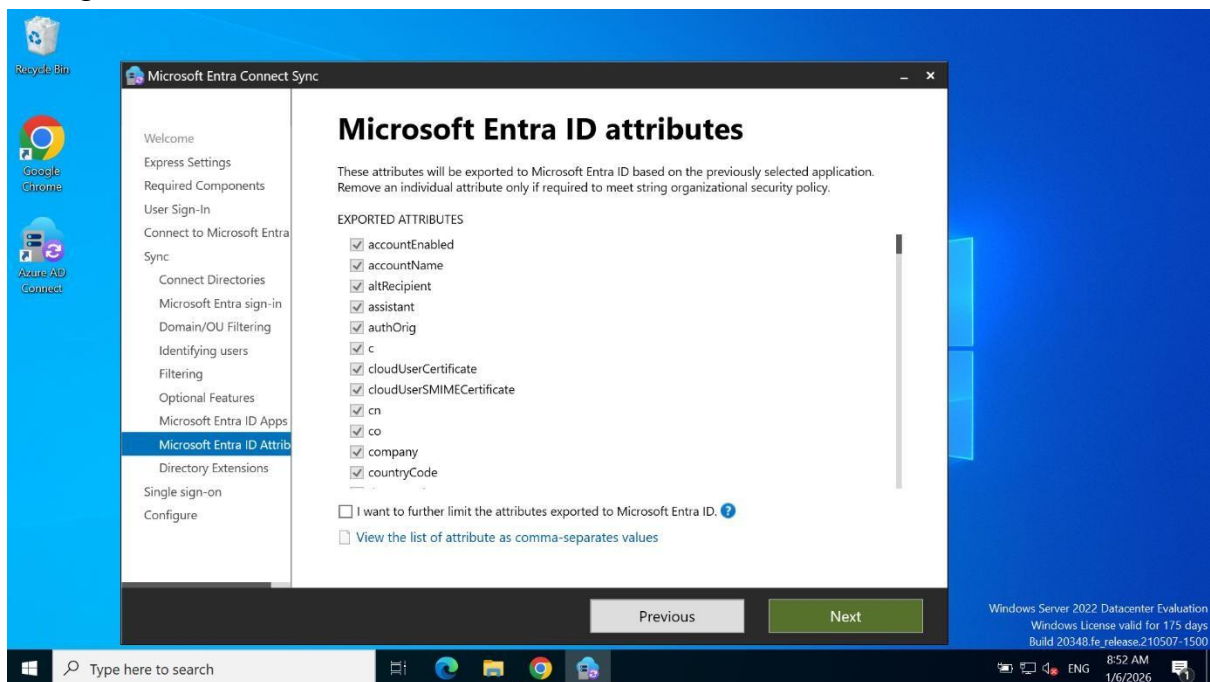
Step 17: Review Microsoft Entra ID Applications

Default Microsoft Entra ID applications such as Office 365, Exchange Online, SharePoint Online, and Intune were reviewed.



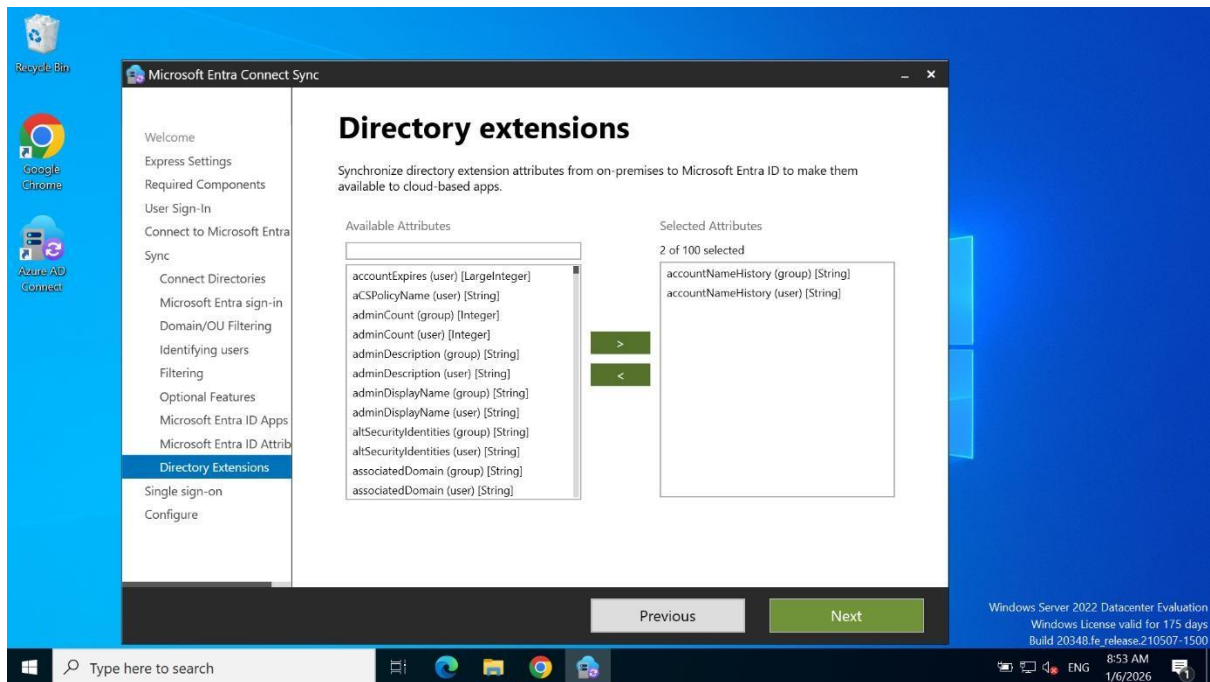
Step 18: Review Attribute Synchronization

The list of attributes to be exported to Microsoft Entra ID was reviewed and left at default settings.



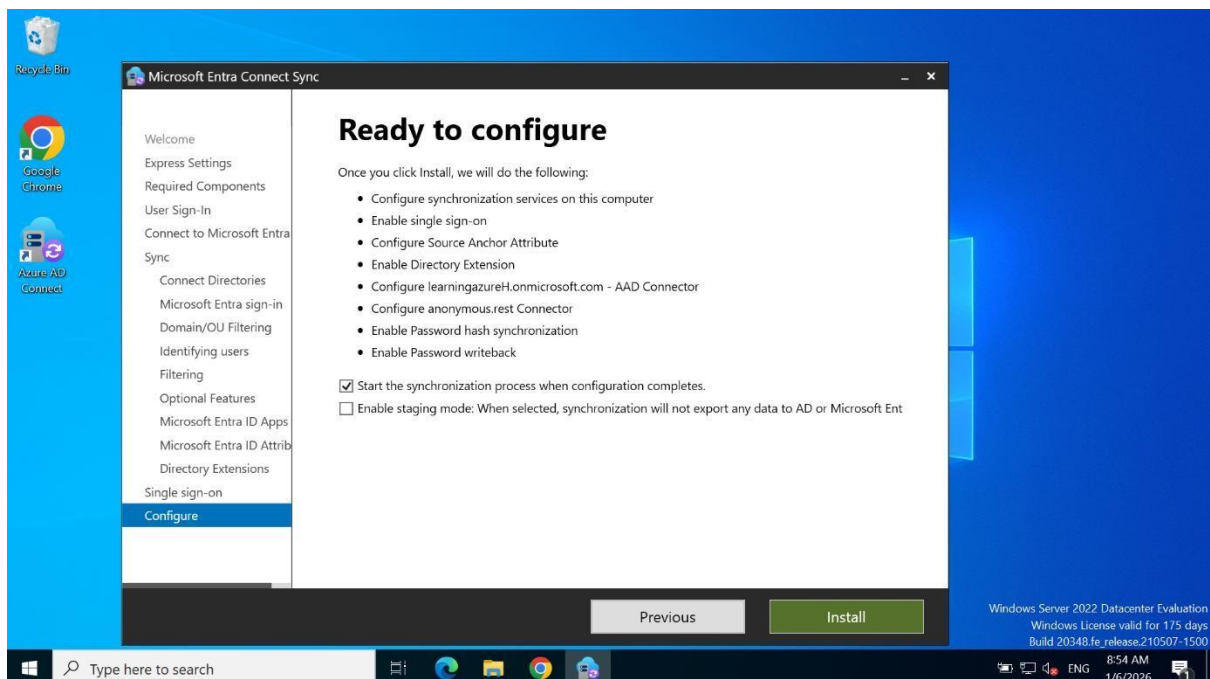
Step 19: Directory Extensions Review

Selected directory extension attributes were reviewed to ensure correct data synchronization.



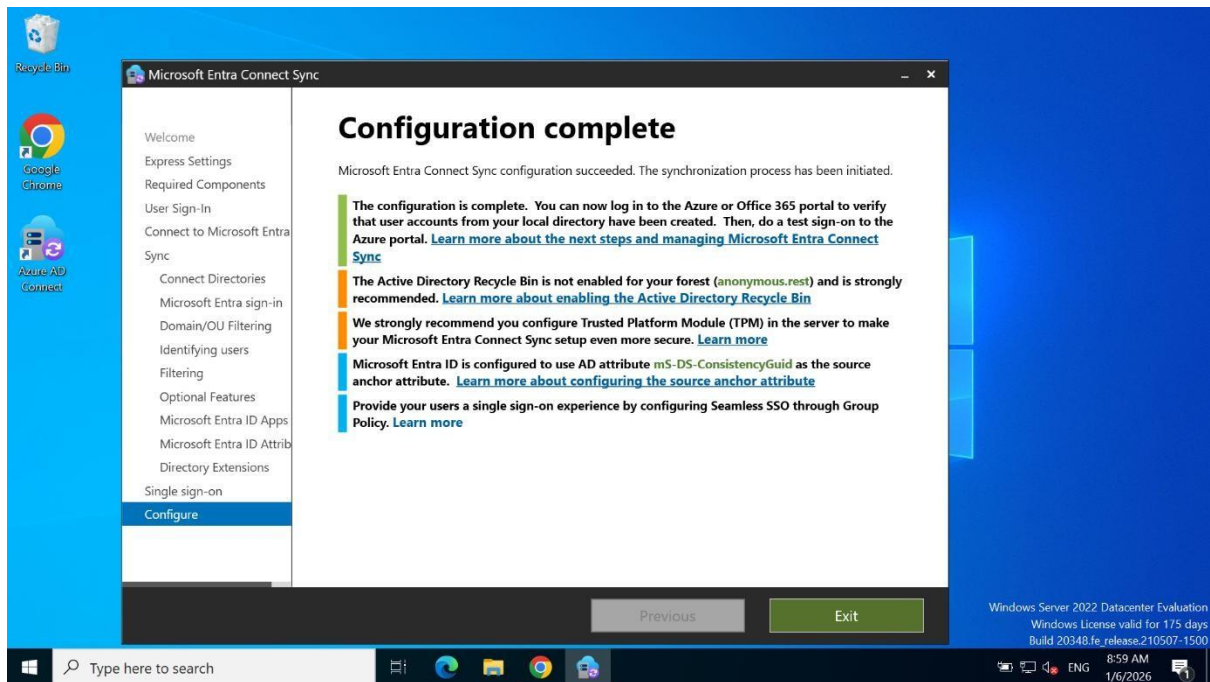
Step 20: Ready to Configure Summary

The configuration summary was reviewed, including connectors, authentication method, and enabled features.



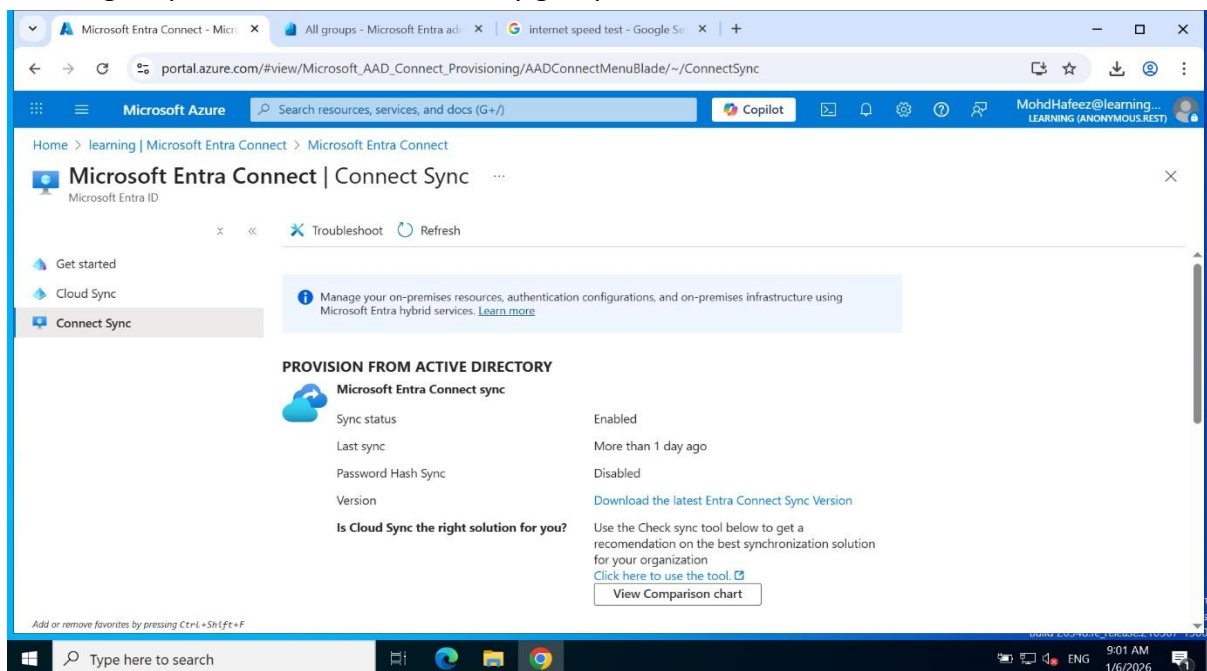
Step 21: Configuration Completion

Microsoft Entra Connect completed configuration successfully and initiated the synchronization process.



Step 22: Validation in Azure Portal

Synchronization was validated by confirming synced users and groups in Microsoft Entra ID, including on-premises users and security groups.



Microsoft Azure portal interface showing the 'Users' management page. The breadcrumb navigation is 'Home > learning | Users > Users'. The page title is 'Users' with a sub-header 'learning'. The left sidebar contains a list of navigation options: All users, Audit logs, Sign-in logs, Diagnose and solve problems, Deleted users, Password reset, User settings, Bulk operation results, Bulk operation results (Preview), and New support request. The main content area displays a table of users with columns: Display name, User principal name, User type, Is Agent, On-premises sy..., and Identities. Three users are listed: 365 user, Mohd Hafeez, and muzammil khan. The table also includes checkboxes for each user. The bottom of the page shows the Windows taskbar with the search bar and system tray.

Display name	User principal name	User type	Is Agent	On-premises sy...	Identities
365 user	365user@learning...	Member	No	No	learningazureH.onm
Mohd Hafeez	MohdHafeez@lea...	Member	No	No	learningazureH.onm
muzammil khan	muzammil.khan@...	Member	No	Yes	learningazureH.onm

Microsoft Azure portal interface showing the 'User Profile' page for 'muzammil khan'. The breadcrumb navigation is 'Home > learning | Users > Users > muzammil khan'. The page title is 'muzammil khan' with a sub-header 'User'. The left sidebar contains a list of navigation options: Overview, Audit logs, Sign-in logs, Diagnose and solve problems, Custom security attributes, Assigned roles, Administrative units, Groups, Applications, Licenses, Devices, and Azure role assignments. The main content area displays the user's profile information, including User principal name, Object ID, Created date time, User type, and Identities. The 'My Feed' section shows the user's account status (Enabled) and sign-in history (Last interactive sign-in, Last non-interactive sign-in). The bottom of the page shows the Windows taskbar with the search bar and system tray.

Property	Value
User principal name	muzammil.khan@anonymous.rest
Object ID	949b8a09-d1d5-450d-a402-828e5aa8bc06
Created date time	Jan 6, 2026, 8:58 AM
User type	Member
Identities	learningazureH.onmicrosoft.com

My Feed

- Account status**
 - Enabled
- Sign-ins**
 - Last interactive sign-in: -- --
 - Last non-interactive sign-in: -- --

The first screenshot shows the 'admins-cloud' group overview in the Microsoft Azure portal. The group is of type 'Security' and has 2 direct members. The second screenshot shows the 'Members' page for the same group, displaying a list of two members: 'shaik abdullah' and 'syed abdullah', both of type 'User'.

Group Overview:

Property	Value
Membership type	Assigned
Source	Windows Server AD
Type	Security
Object ID	5ab7977b-b6f8-461c-9d30-95750f317d50
Created on	1/6/2026, 9:28 AM
Total direct members	2
User(s)	2
Group(s)	0
Device(s)	0
Other(s)	0

Members:

Name	Type	Email	User type	Object Id
shaik abdullah	User		Member	024d0c35-134a-4dac-8bc
syed abdullah	User		Member	6884d141-206c-470c-a05

Validation and Verification

- Sync status confirmed as **Enabled** in Azure Portal
- On-premises users visible in Microsoft Entra ID
- Group memberships synchronized correctly
- User identities show on-premises source

Outcome

A hybrid identity environment was successfully implemented. On-premises Active Directory users and groups are now synchronized with Microsoft Entra ID, enabling centralized identity management and seamless access to cloud services.

Skills Demonstrated

- Hybrid Identity Implementation
 - Microsoft Entra Connect Configuration
 - Active Directory to Entra ID Synchronization
 - OU-based Filtering
 - User and Group Validation
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Subscription Used

Microsoft Azure Subscription (Learning / Pay-As-You-Go)